Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

ASEA

Selected Domain

Domain 1: The Aiversecity Future-Ready Workforce

Selected SDG Track (1–17)

SDG 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17

Project / Solution Name

Path.AI: find your path in the AI era

Name (Leader)

Almat Sakiidulla

Email & Phone (Leader)

sat200707@mail.ru, +77072203626(WhatsApp)

University Affiliation(s)

Xiamen University Malaysia

Number of Team Members

4

GitHub Repository github.com/Almatikooo/Competition_Repo

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Artificial intelligence is reshaping the labour market, transforming occupations and creating demand for new skills. Many people struggle to understand these changes, identify sustainable career directions, and determine what they need to learn.

Evidence from international online work illustrates this challenge. Demirci, Hannane, and Zhu (2024) found that job postings for automation-prone writing and coding tasks declined by 21% relative to jobs requiring manual-intensive skills within eight months of ChatGPT's introduction on a leading global freelancing platform. Although this finding does not represent the entire global labour market, it demonstrates reduced opportunities in specific occupations exposed to generative AI.

Our target groups include school leavers, university graduates, unemployed adults, and workers whose roles may be transformed or displaced. Experienced individuals need to identify transferable skills and transitions, while people with limited skills need a clear starting point for learning and employment.

Without accessible, evidence-based guidance, individuals risk spending time and money on unsuitable training, overlooking opportunities, or pursuing skills that do not match changing requirements. Those with limited resources face additional disadvantages.

The central challenge is connecting evolving labour market signals with individual abilities and learning needs, enabling informed career decisions, relevant education, lifelong learning, and preparation for decent work.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Path.AI is a web application that translates changing AI and labour market signals into explainable career recommendations and practical, affordable learning pathways.

Users build a skills profile by selecting their occupation or field of study. Skills are drawn from 35 catalogued roles linked to O*NET codes. Demonstrations use synthetic profiles and CVs, avoiding real personal data.

The platform combines two dated sources: approximately 20,000 job postings across 36 Hacker News hiring threads and OpenAlex research activity covering 12 AI capabilities. These signals inform modelled demand estimates for 44 skills under slower, expected, and faster AI adoption scenarios. Estimates include ranges and assumptions rather than promises of future employment.

An algorithm ranks roles using skill fit, transferable strengths, outlook, interests, and SDG preferences. Recommendation cards explain the match, estimate learning gaps in hours, and show whether a role remains suitable across scenarios.

Personalised pathways adapt to weekly study availability and combine curated learning materials, verified free or affordable courses, and links to Malaysian employment and training resources. Quizzes support progress tracking; completion certificates represent platform achievements, not accredited qualifications.

Every recommended role and skill maps to specific SDG targets with an explanation, making sustainability part of career exploration rather than a decorative label.

We apply workshop topics through GenAI API integration, validated database architecture, dated source caching, and cloud deployment on Replit and RunPod. An open-weight model supports skill extraction and readable explanations; deterministic logic controls numerical scores and catalogue selection. Guardrails, source checks, and human review keep recommendations transparent and accountable.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Path.AI will use a modular architecture that separates data collection, numerical calculations, AI-assisted interpretation, and the user interface.

Frontend. React, TypeScript, Tailwind CSS: Responsive interface for profiles, career recommendations, scenario comparisons, and learning pathways

Backend. Python, FastAPI: API endpoints, input validation, source processing, matching, and pathway generation

Database. PostgreSQL: Structured storage for roles, skills, SDG mappings, learning resources, source snapshots, and recommendation results

Labour market data. Hacker News API: Retrieve public hiring threads and comments for processing into dated job-posting observations

AI research signals. OpenAlex API: Retrieve publication metadata associated with selected AI capabilities

AI inference. An appropriately licensed Qwen-family model served through vLLM: Skill extraction, terminology normalisation, and readable explanations

GPU infrastructure. RunPod: Host the model inference service

Application hosting. Replit: Develop, run, test, and publish the application

Version control. GitHub: Maintain source code, documentation, licensing information, and development history

Continued: architecture notes and block diagram (4a); AI model, APIs, data sources and hosting (4b).

This architecture applies to the workshop topics of GenAI API integration, database architecture, and cloud deployment. The backend will communicate with vLLM through its OpenAI-compatible API; this interface does not imply that OpenAI hosts the model.

Official SDC Sponsor-Provided Resources

Our planned architecture incorporates two official SDC sponsor platforms:

- **Replit:** application development, testing, backend execution, and publication of the functional prototype.
- **RunPod:** GPU infrastructure for model inference through the vLLM endpoint.

Actual usage and any sponsor credits consumed will be recorded in the Technical & Tools Declaration, alongside the required evidence. The declaration will distinguish resources genuinely provided through the competition from independently purchased services.

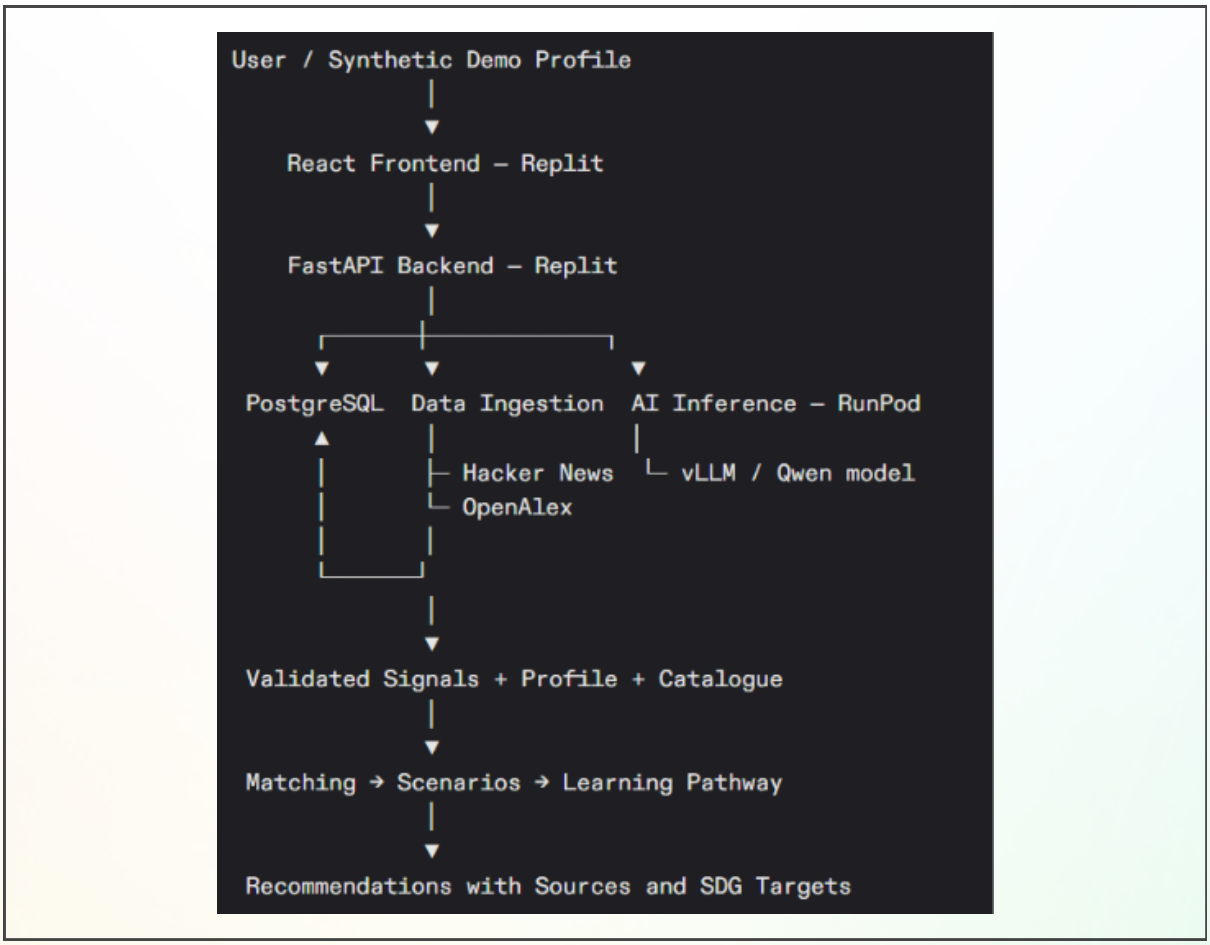
3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

3.1 (continued). Architecture notes and System Block Diagram

The backend will validate incoming records and store dated snapshots for reproducible comparisons. Deterministic application logic will calculate skill fit, scenario-based estimates, learning gaps, and catalogue selection. The language model will assist with interpretation and explanation rather than independently inventing jobs, courses, or numerical forecasts.

Learning resources will come from a curated catalogue. Quiz grading and certificate signing will run on the server, with credentials and signing secrets excluding from the public repository.



3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

3.1 (continued). AI model, APIs, data sources and hosting

AI model

- Qwen3.8-27B-FP8 (Qwen team, Alibaba; open weights, Apache-2.0), self-hosted with vLLM on a RunPod RTX A6000 GPU.
- Backup while our GPU is off: Groq API serving the same open model (qwen/qwen3.8-27b). It receives only the facts behind an answer, never a CV.
- Every AI answer is grounded in facts Path.AI computes. Numbers not found in those facts are rejected.

APIs used

- **Groq API:** hosted backup for the language model
- **RunPod REST API:** switching the GPU on and off
- **Hacker News Search API (Algolia):** live trend signal from "Who is hiring?" posts
- **OpenAlex API:** live trend signal from AI research
- **Greenhouse, Lever, Ashby and Workable public job-board APIs:** company job postings
- **Himalayas, Jobicy, Remotive, Remote OK and Arbeitnow APIs:** remote jobs
- **Wikidata (QLever SPARQL), Hipo university-domains-list, GeoNames:** universities and locations

Data sources (cited)

- WEF Future of Jobs Report 2025 (incl. the Malaysia profile) and 2023, ILO Working Paper 140, O*NET, UN SDGs, Hays 2025 press release
- Open-licensed courses (OpenStax, Microsoft "for Beginners" and others; no NC licences)

Hosting

- DigitalOcean VPS (Singapore), Caddy + Let's Encrypt

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We applied design thinking to translate career uncertainty into an accessible, feasible experience. During the empathise stage, we created five fictional personas: Aisyah, 34, an accounts clerk facing automated invoice processing; Wei Jie, 23, an electrical engineering graduate choosing his first direction; Kavitha, 41, a customer-service lead whose contact centre is introducing chatbots; Hafiz, 29, a graphic designer affected by generative AI; and Nurul, 27, an agricultural extension officer exploring agritech.

These personas helped us define one question: “As AI changes my opportunities, what should I learn next?” Their different backgrounds informed editable profiles, transferable-skill recognition, and learning pathways adapted to available time and budget.

During ideation, we rejected generic chatbot advice. Dated signals support scenario estimates, deterministic logic calculates recommendations, and the language model assists interpretation and explanation.

We prototyped a short journey with selectable examples, prefilled skills, and expandable reasoning. Automated tests cover matching, guardrails, fairness, and learning functionality. Phase 2 will add timed peer usability sessions to identify confusing steps and improve accessibility.

Replit hosts the application, while RunPod supports optional GPU inference. Incremental updates preserve the last signal cache. Deterministic fallbacks maintain core functionality during model outages, supporting reliable demonstrations and preparation for Phase 2 deployment.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

Path.AI commits to three measurable KPIs, one per SDG sub-target, each tested in Phase 2 with at least 20 peer testers using fictional or self-created profiles only.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary — SDG 4.4: Increase the number of young people and adults with relevant skills for employment, decent jobs, and entrepreneurship. Path.AI supports this through transferable-skill identification, skills-gap analysis, and personalised reskilling pathways. Supporting — SDG 4.3: Promote access to affordable technical, vocational, and tertiary education through verified free or low-cost learning resources. Supporting — SDG 8.5: Support productive employment and decent work through informed career exploration and preparation for occupational transitions. Supporting — SDG 8.2: Support productive employment and decent work through technological upgrading and innovation. The prototype measures career-planning efficiency and accessibility, not guaranteed employment outcomes.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Demirci, Hannane, and Zhu (2024) found a 21% relative decline in job postings for automation-prone writing and coding work on a leading global freelancing platform within eight months of ChatGPT's introduction. This demonstrates disruption in a specific market, not a worldwide decline in employment. For prototype evaluation, our provisional, unvalidated planning estimate is 30 minutes to manually identify a suitable role, check skill requirements, and assemble a sourced learning pathway. Timed manual tasks using our five fictional personas will establish the measured baseline before Phase 2 evaluation.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	Percentage reduction in median time required to produce a complete, evidence-supported career plan using Path.AI compared with manual research. A complete plan includes a recommended role, suitability explanation, skill gaps, ordered learning pathway, accessible resource links, and SDG connections. Equivalent tasks will use the same five synthetic profiles and completeness checklist. Formula: Time reduction (%) = [(Manual median time − Path.AI median time) ÷ Manual median time] × 100. Source validity and completeness will be checked alongside speed.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Achieve at least a 50% reduction in median career-planning time relative to the measured manual baseline by the final prototype presentation. Using the provisional 30-minute estimate, this corresponds to 15 minutes or less. The absolute threshold will be updated after baseline measurement. Additional targets: 100% completion of required plan elements and 100% valid supporting references and learning-resource links across the five benchmark profiles. These are proposed targets, not achieved results. Evaluation will report the sample size, observed outcomes, and limitations without claiming proven employment or income improvements.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

No. The repository was created on 19 October 2026 at 22:55 MYT, after the problem statements were released at 22:00. All Path.AI code, catalogs, scoring models and lesson notes were written during the hackathon.

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
FastAPI Uvicorn Pydantic	Code (web framework) Code (ASGI server) Code (data validation)	MIT BSD-3-Clause MIT
httpx python-multipart pypdf pytest	Code (HTTP client) Code (file upload) Code (CV PDF text) Code (tests)	BSD-3-Clause Apache-2.0 BSD-3-Clause MIT
Roboto (Google Fonts) vLLM OpenAlex works API	Code (font) Code (model server) Dataset (AI research volume)	SIL OFL 1.1 Apache-2.0 CC0
HN "Who is hiring?" threads via HN Algolia API Learn lesson sources (The Carpentries and other open courses, UK and US government material, Wikipedia)	Dataset (public job posts, aggregate counts only) Dataset (lesson readings)	Public API; Y Combinator terms CC BY 4.0, MIT, OGL v3.0, US public domain, CC0, CC BY-SA 4.0
Qwen3.8-27B-FP8 (Alibaba Qwen) O*NET-SOC occupation codes WEF Future of Jobs 2023 and 2025	Model, pre-trained, served via vLLM Dataset (role catalog) Dataset (cited baseline claims)	Apache-2.0 CC BY 4.0 Cited with attribution

Yes. All are open-source or open data, declared and attributed in the repository.

AI-assisted development tools are declared separately in the Technical & Tools Declaration, section 4.3.

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]
Almat Sakiidulla

[Date]
24 October 2026